

Practice Set

A. Arithmetic Operations

- ~~1. Calculate the sum of the first n natural numbers. Take n as input.~~
- ~~2. Take two numbers as input and perform addition, subtraction, and multiplication.~~
- ~~3. Take two numbers a and b and calculate:~~
 - ~~⊖ Integer division of a by b~~
 - ~~⊖ Floating-point division of a by b~~
 - ~~⊖ Remainder of a divided by b~~
- ~~4. Take two numbers a and b and calculate a^b .~~
- ~~5. Find the difference between the square of the sum and the sum of the squares of the first n natural numbers.~~
- ~~6. Take three single-digit numbers x , y , z and print them as a single number xyz using arithmetic operations (no string concatenation).~~

B. Geometry & Measurement

- ~~1. Calculate the area and circumference of a circle. Take the radius as input.~~
- ~~2. Calculate the area of a triangle. Take base and height as input.~~
- ~~3. Take four integers x_1 , y_1 , x_2 , y_2 and calculate the Euclidean distance between the points (x_1, y_1) and (x_2, y_2) . Round the result to two decimal places.~~

C. Temperature Conversion

- ~~1. Convert a temperature from Celsius to Fahrenheit. Take the Celsius temperature as input.~~
- ~~2. Convert a temperature from Fahrenheit to Celsius. Take the Fahrenheit temperature as input.~~

D. Quadratic Equations

- ~~1. Compute the roots of a quadratic equation $ax^2 + bx + c = 0$, assuming the roots are real and distinct. Take a , b , c as input.~~

E. Number Manipulations

- ~~1. Take a number x and output $x + xx + xxx + xxxx$.~~
- ~~2. Calculate the cube of a number. The number should be user input.~~
- ~~3. Swap two numbers x and y and print the swapped values.~~

- ~~4. Take a three-digit number and print each digit on a separate line, starting from the unit's place.~~

F. Conditional & Logical Checks

1. Check whether a character is an alphabet (a–z or A–Z). Print 1 if it is, otherwise print 0.
2. Check whether a year is a leap year. Print 1 if it is, otherwise print 0.

G. Additional Questions

1. Write a C program that takes four integers as input from the user and prints the second largest number among them.

Conditions/Restrictions:

- I. You are not allowed to use loops (for, while, do-while).
- II. You are not allowed to use arrays or any data structures to store the numbers.
- III. You must solve the problem using only nested if-else statements to compare the numbers.

2. Write a program that takes time input in 24-hour format as two integers:

- hh (hours, 0–23)
- mm (minutes, 0–59)

Tasks:

- If hh or mm are outside their valid ranges, print "Invalid Time".
- Otherwise, classify the time:
 - Morning: 05:00–11:59
 - Afternoon: 12:00–16:59
 - Evening: 17:00–20:59
 - Night: 21:00–23:59 and 00:00–04:59

3. Write a program that takes three integers as input, representing day, month, and year. Your task is to determine whether these values form a valid calendar date.

Requirements:

- I. Validate that the month is between 1 and 12.
- II. Validate that the day is within the correct range for the given month:

- A. Months with 31 days → January, March, May, July, August, October, December.
 - B. Months with 30 days → April, June, September, November.
 - C. February → 28 days in a common year, 29 days in a leap year.
- III. A leap year is defined as:
- A. divisible by 400, OR
 - B. divisible by 4 but not by 100.
- IV. If the date is valid, print "Valid Date". Otherwise, print "Invalid Date".

Example Runs:

- Input: 29 2 2024 → Output: Valid Date (2024 is a leap year)
- Input: 29 2 2023 → Output: Invalid Date (2023 is not a leap year)
- Input: 31 4 2022 → Output: Invalid Date (April has only 30 days)